

internal-external rotation 90 degree-abducted — ex vivo – in vivo difference, 95% CI (overlap only)

Sternoclavicular — Protraction(+)/retraction(-)

Sternoclavicular — Depression(+)/elevation(-)

Sternoclavicular — Backwards(+)/forward(-) rot

not modelled

not modelled

not modelled

Acromioclavicular — Protraction(+)/retraction(-)

Acromioclavicular — Medial(+)/lateral(-) rot

Acromioclavicular — Posterior(+)/anterior(-) tilt

not modelled

not modelled

not modelled

Scapulothoracic — Protraction(+)/retraction(-)

Scapulothoracic — Medial(+)/lateral(-) rot

Scapulothoracic — Posterior(+)/anterior(-) tilt

single group $\cdot K = 7$

single group · K = 5

single group · $K = 6$

Glenohumeral — Plane of elevation

Glenohumeral — Elevation(-)/Depression(+)

Glenohumeral — Int(+)/ext(-) rotation

single group · $K = 8$

single group · K = 5

single group · K = 6

blue line = EX VIVO – IN VIVO, band = 95% CI · squares along the bottom = band excludes 0. Note the sign: in vivo is the reference, so this is mirrored against v2.
stars = LIKELIHOOD-RATIO TEST of H0 'no ex-vivo departure' against H1, Bonferroni-adjusted across every compared cell on the plate: *** p<0.001, ** p<0.01, * p<0.05, ns = not significant.

ex/in · studies ex/in | K, this cell's own spline df chosen by BIC | mean and max $|\Delta|$ in degrees. Curves span the full abscissa — every x any shoulder reaches is fitted, so the thinnest stretches rest on a single shoulder and the 95%